

Crop Labeller — Researcher Guide

This guide is for reviewing wheat NDVI labels — no coding or "developer" background needed. You already know wheat when you see its curve; this just walks you through setting up and then the screen, click by click.

Everything you do in the app — picking your file, looking at data, saving your work — happens by clicking in your web browser, same as any website. On Windows you start it by double-clicking a file; nothing needs typing.

1. Pick your setup

Choose the **first row that fits your computer**. The rows go from easiest to most involved.

Your computer	Use	What it takes
Windows (with or without internet)	Option A — Ready-made package (recommended)	Unzip, then double-click. Nothing to install; Python is built in.
Windows that must use its own Python 3.12	Option B — Your own Python	Unzip, then double-click an installer once.
Mac , or you already use <code>uv</code>	Option C — Install with <code>uv</code>	Needs internet. Install <code>uv</code> , then type one command.
Any computer , if you're comfortable with Python and prefer no ready-made zip	Option D — From the source code, your own way	Get the code, make a virtual environment, <code>pip install -r requirements.txt</code> , <code>python run.py</code> .

Options A and B work on computers with **no internet at all**. You only need internet somewhere to download the zip, which you can then copy over, e.g. on a USB drive. Option D can work offline too, if you bring your own package files.

Are the zips safe? Yes. GitHub builds them automatically from this project's public source code, and each one is tested on Windows before it is published. They contain only the app, a standard Python (the same builds the `uv` tool uses), and the usual open-source packages from PyPI (Streamlit, pandas, Plotly and their dependencies). `SHA256SUMS.txt` on the download page lets you check a copy wasn't changed on the way. If your IT rules still don't allow running a downloaded package, use **Option D**: you install everything yourself from the source code.

2. One-time setup

Do only the option you picked above.

Option A — Ready-made package (Windows, recommended)

1. **Download** `crop-labeller-<version>-windows-offline.zip` from <https://github.com/vickvey/crop-labeller/releases/latest> (under **Assets**), or get it from whoever gave you this tool. If the computer you'll use has no internet, download it elsewhere and copy it over.
2. **Unblock it.** Right-click the zip → **Properties** → at the bottom, tick **Unblock** → **OK**. (If there's no Unblock box, skip this.) This stops Windows from showing warnings about the files inside.
3. **Extract it.** Right-click the zip → **Extract All...** → **Extract**. The suggested location is fine. You'll get a folder containing `crop-labeller`. Move that `crop-labeller` folder wherever you like, for example your Documents.

Don't double-click files *inside* the zip without extracting it first. It won't work from there.

That's all the setup. Go to [Section 3](#).

Option B — Your own Python 3.12 (Windows)

For computers that must use the Python already installed on them. It must be **Python 3.12, 64-bit**.

1. Do steps 1–3 of [Option A](#), but download `crop-labeller-<version>-windows-offline-own-python.zip` instead.
2. In the extracted `crop-labeller` folder, double-click `install-with-own-python.bat`. It finds your Python 3.12, creates a private `.venv` folder, and installs the included packages from the `wheelhouse` folder, offline. When it says **Setup finished**, press a key to close it. You only do this once, or again if you switch to a different Python.

That's all the setup. Go to [Section 3](#).

If it says it couldn't find a 64-bit Python 3.12, tell it exactly which Python to use:

1. Open PowerShell in the `crop-labeller` folder. Hold **Shift**, **right-click** empty space inside the folder, and choose **Open PowerShell window here** (on Windows 11: **Open in Terminal**).
2. Run the installer with the full path to your `python.exe`, e.g.:

```
.\install-with-own-python.bat  
C:\Users\you\AppData\Local\Programs\Python\Python312\python.exe
```

To find the path, run `py -3.12 -c "import sys; print(sys.executable)"`, or look at where Python was installed.

Prefer to install by hand? These are the same steps as the installer, run in PowerShell in the `crop-labeller` folder:

```
py -3.12 -m venv .venv
.venv\Scripts\python -m pip install --no-index --find-links wheelhouse -r
requirements.txt
.venv\Scripts\python run.py
```

`requirements.txt` pins the exact versions that are in `wheelhouse` and that the app is tested with.

- **Using wheels you downloaded yourself** instead of the included `wheelhouse`? Point `--find-links` at your folder, and install by name so pip picks versions from what you have:
`.venv\Scripts\python -m pip install --no-index --find-links C:\path\to\your\wheels streamlit pandas plotly`. The app needs Streamlit ≥ 1.38 , pandas ≥ 2.2 and Plotly ≥ 5.24 .
- **Using uv** (already installed offline)? Run `uv venv --python C:\path\to\python.exe`, then `uv pip install --offline --no-index --find-links wheelhouse -r requirements.txt`, then `.venv\Scripts\python run.py`. (Plain `uv run run.py` won't work offline: it tries to download packages.)

Option C — Install with `uv` (Mac, or any computer with internet)

`uv` is a small helper program that sets up everything else automatically, so you don't need to install Python separately. It needs internet.

1. **Install `uv`**. Open a terminal:

- **Mac:** open **Terminal** (search for it with Spotlight, `Cmd+Space`).
- **Windows:** press the Start button, type `PowerShell`, and open **Windows PowerShell**.

Copy-paste the line for your computer and press **Enter**:

Mac:

```
curl -Lsf https://astral.sh/uv/install.sh | sh
```

Windows (PowerShell):

```
powershell -ExecutionPolicy ByPass -c "irm
https://astral.sh/uv/install.ps1 | iex"
```

It takes a few seconds. You won't need to do this again on this computer.

2. **Get the project folder.** Download it as a `.zip` (from the link you were given, or **Code** → **Download ZIP** on the GitHub page). Unzip it: double-click it on a Mac, or right-click → **Extract All** on Windows. You'll get a folder named something like `crop-labeller`. Put it somewhere easy to find, like your Desktop.

That's all the setup. Go to [Section 3](#).

Option D — From the source code, your own way (any computer)

The traditional Python way, with no ready-made package. You need **Python 3.10 or newer** (64-bit), and either internet access for `pip` or your own package files (see "Offline" below). Windows commands are shown first, then Mac/Linux.

1. **Get the code**, whichever way you like:

- `git clone https://github.com/vickvey/crop-labeller.git`
- or **Code** → **Download ZIP** on <https://github.com/vickvey/crop-labeller>
- or **Source code (zip)** under a release on <https://github.com/vickvey/crop-labeller/releases>

Then open a terminal in the `crop-labeller` folder, e.g. `cd crop-labeller`.

2. **Create a virtual environment** (or use any environment you prefer):

```
py -m venv .venv          # Windows
python3 -m venv .venv     # Mac / Linux
```

3. **Install the packages:**

```
.venv\Scripts\python -m pip install -r requirements.txt    # Windows
.venv/bin/python -m pip install -r requirements.txt        # Mac / Linux
```

`requirements.txt` pins the exact versions the app is tested with, for Python 3.10 and newer. If you prefer to activate the environment first (`.venv\Scripts\activate` on Windows, `source .venv/bin/activate` on Mac/Linux), plain `python` and `pip` work too.

That's all the setup. Go to [Section 3](#).

Offline, with your own package files: download the wheels (`.whl` files) on a computer with internet. Run `pip download -r requirements.txt -d wheels` there. If that computer's Python version or operating system differs from the offline one, add e.g. `--only-binary=:all: --python-version 3.12 --platform win_amd64` to match it. Copy the `wheels` folder over, then install with `pip install --no-index --find-links wheels -r requirements.txt`. If your wheels are different versions from the pinned ones, install by name instead: `pip install --no-index --find-links wheels streamlit pandas plotly`.

Your own tools are fine. conda, Poetry, `uv pip`, a shared lab environment and so on all work. The app only needs `streamlit ≥ 1.38`, `pandas ≥ 2.2` and `plotly ≥ 5.24` installed. Then run `python run.py` from the `crop-labeller` folder. There's nothing to install for the app itself.

3. Every time you want to review data

Step 1 — Put your CSV file in the `data` folder

1. Open the `crop-labeller` folder.
2. Open the `data` folder inside it.
3. Drag and drop your assigned CSV file (the one with the datapoints you need to review) into this `data` folder — just like copying any file.

That's the only file-handling step. You never need to open or edit the CSV yourself.

Step 2 — Start the app

Option A or B (Windows package): double-click `start-crop-labeller.bat` in the `crop-labeller` folder. A black window opens.

Option C (`uv`): open a terminal in the `crop-labeller` folder (not inside `data`):

- **Mac:** right-click the `crop-labeller` folder and choose **New Terminal at Folder** (or open Terminal, drag the folder into it, and press Enter).
- **Windows:** inside the `crop-labeller` folder, hold **Shift**, **right-click** empty space, and choose **Open PowerShell window here** (on Windows 11: **Open in Terminal**).

Then paste this and press **Enter**:

```
uv run run.py
```

The first time, this takes a minute or two while it installs everything the app needs. After that it starts in a few seconds.

Option D (source code): in a terminal in the `crop-labeller` folder, run the app with your environment's Python:

```
.venv\Scripts\python run.py      # Windows  
.venv/bin/python run.py         # Mac / Linux
```

(or just `python run.py` if the environment is activated).

Either way, a browser tab opens with the app a few seconds later. If it doesn't, look in the black/terminal window for a line like `URL: http://localhost:8501` and paste that address into your browser.

Leave that window open while you work. Closing it stops the app. You can minimize it and ignore it otherwise.

4. Understanding the screen

Top of the page

- 🌾 **Crop Labeller** — the app name, top left.
- 📄 **Working on <filename>** — top right, confirms which CSV you're currently reviewing (useful if more than one file is in `data/`).
- If more than one CSV is in `data/`, a dropdown in the left sidebar lets you pick which one to work on.

Left sidebar

- **Progress** — how many rows you've reviewed out of the total, as a count, a percentage, and a progress bar. This is saved automatically — closing and reopening the app later picks up exactly where you left off.
- **This file** — a quick summary of what's in the CSV: how many rows are currently labelled wheat vs. non-wheat, and (if available) how many rows an automated check flagged as possibly mislabelled.

The two filter rows

- **Show:** `All` / `Unreviewed` / `Reviewed`. Switch to **Unreviewed** to only step through rows you haven't gotten to yet — the most useful setting for a normal review session.
- 🎯 **Confidence filter** (only appears if the file has confidence scoring): click it to open a small panel where you can:
 - Turn the filter on/off.
 - Choose whether to look at the **wheat** rows or the **non-wheat** rows.
 - Drag a slider to pick a percentage (e.g. 10%) — this shows only the rows whose "how well does this match a typical curve" score is in the bottom N% for that group, i.e. the most suspicious/least confident ones. A live count tells you how many rows that is.

This is a good way to triage: turn it on, start with a low percentage like 5–10%, and work through the most questionable rows first before moving to a general pass.

Whatever's on the right side of the top row (e.g. 4454 / 9934 · sample 1695 · Not yet reviewed) tells you exactly where you are and, if a filter is active, your position within the filtered set too.

The main plot

For the row currently on screen, you'll see a chart with the growing season along the bottom (October through May, in half-month steps) and NDVI values (a measure of how "green"/vegetated the ground looked from satellite) up the side. Up to three lines can appear:

- **Original** (solid, with dots) — the row's actual recorded values. This is the real data point you're judging.
- **Smoothed** (thinner blue line, if available) — the same curve with noise smoothed out, useful for seeing the overall shape more clearly.
- **Regional mean** + a shaded band (if available) — what a *typical* wheat curve looks like for this region and season, $\pm$ one standard deviation. Compare your row's curve against this band: a wheat-labelled row that looks nothing like it, or a non-wheat row that matches it closely, is worth a second look. A caption under the chart tells you how many samples that reference curve is based on — treat a reference built from very few samples with a bit more caution.

Hover your mouse anywhere on the chart to see the exact values for that time period across all the lines at once, with a line marking exactly where you're pointing.

Below the chart, an **Other fields** section (click to expand) shows any extra columns from the CSV — coordinates, region, cover area, and so on — for additional context if you need it.

The review panel (right side)

- **Original label** — what the row is currently labelled as in the file.
- **Label confidence** (if available) — a colored badge (green/orange/red) showing how well an automated check thinks the current label matches the curve's shape. Low confidence doesn't mean it's wrong, just that it's worth a closer look.
- A **warning banner** (if available) explains specifically what looks off, e.g. *"Labeled non-wheat, but its NDVI shape looks like a typical wheat curve."*
- **Label** — click **wheat** or **non-wheat** to set what you believe the correct label is. The whole page background tints a very pale green or red to match your current choice — a quick visual double-check before you save.
- **Comment** (optional) — a text box for notes, e.g. why you think a label is wrong, or anything unusual about the curve.
- If you've changed something but haven't clicked Save yet, a small  **Unsaved change** note appears as a reminder.
- **Prev / Next** — move to the previous/next row (respecting whatever Show/Confidence filters are active).
- **Save** — saves your label and comment for this row, and automatically moves to the next row.

- **Jump to row** — type a row number to go straight to it.
-

5. Reviewing a row, step by step

1. Look at the **Original** curve and, if shown, the **Regional mean** band and **Smoothed** curve.
 2. Check the **Original label** and, if shown, the **confidence badge** and **warning banner**.
 3. Decide: does the curve genuinely look like wheat, or not?
 4. Click **wheat** or **non-wheat** under **Label** (leave it unchanged if you agree with the original).
 5. Optionally, type a note in **Comment** — especially useful for borderline or unusual cases, so whoever looks at this later understands your reasoning.
 6. Click **Save**. You'll move automatically to the next row (unless you're already on the last row matching your current filters, in which case you'll stay put — your save still went through).
 7. Repeat. Use the **Show** and **Confidence filter** controls any time to change which rows you're stepping through.
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6. Finishing up

Your work is saved continuously — every click of **Save** writes to disk immediately, so there's no separate "submit" step and nothing is lost if you close the app partway through.

When you're done (or want to hand off progress), look in the `output` folder inside `crop-labeller`. For a file named e.g. `MyFile.csv`, you'll find:

- `MyFile_labelled.csv` — the full dataset with your corrected labels.
- `MyFile_review_meta.csv` — a log of every row you reviewed: the original label, your label, your comment, and when you reviewed it.

Send both of these files back (however you were asked to — email, shared drive, etc.). Your original input CSV in `data/` is never changed.

7. Stopping the app

Close the black/terminal window (or click into it and press **Ctrl+C**). To work again later, repeat [Section 3](#). Your progress will still be there.

8. Troubleshooting

Setting up and starting

"Windows protected your PC" when you double-click a `.bat` → click **More info** → **Run anyway**. (Unblocking the zip before extracting, in Option A step 2, prevents this.)

"Can't find python\python.exe next to this file" → you ran it from inside the zip, or only copied some of the files. Extract the whole zip and run the `.bat` inside the extracted `crop-labeller` folder.

"Setup hasn't been run yet" (Option B) → double-click `install-with-own-python.bat` first.

The install stops with an error about a package or version (Option B) → your Python must be **3.12, 64-bit**. Check with `py -3.12 -c "import sys, struct; print(sys.version, struct.calcsize('P') * 8, 'bit')"`. If it's a different version, use Option A instead. It has its own Python.

Extracting fails with "path too long" or "file name too long" → extract to a shorter place instead, e.g. type `C:\cl` as the destination in **Extract All**.

The browser didn't open automatically → copy the `http://localhost:8501` -style address printed in the black/terminal window and paste it into your browser's address bar.

Which version do I have? (Options A and B) Open `VERSION.txt` in the `crop-labeller` folder, and include it if you report a problem.

While using the app

"No CSV files found in data" → you haven't put a CSV file directly in the `data` folder yet (not in a subfolder). Add it and reload the page (F5 in your browser).

"Couldn't read <file> ..." → the file isn't in the expected format. Double check it's the CSV you were given, and that it wasn't accidentally edited or renamed. If you have more than one CSV in `data/`, make sure the right one is selected in the sidebar.

I closed the black/terminal window by accident → no problem, nothing is lost. Start the app again ([Section 3, Step 2](#)); your saved progress is still there.

The app looks broken/stuck → reload the page (F5). If that doesn't help, close the black/terminal window and start the app again ([Section 3, Step 2](#)).

I want to review a different file → either remove the current CSV from `data/` and add the new one, or keep both in `data/` and pick the one you want from the dropdown that appears in the sidebar.

Questions or something not working as described here? Reach out to whoever gave you this tool.